

Simple Regime Rules, Not More Factors: A Historical Case Study in Factor-Curve Portfolio Construction

Evidence from U.S. Equity Research Panel, 2002-2025

Rodrigo Sucupira Andrade de Carvalho Lima

Independent Researcher

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Abstract

This paper studies whether incremental improvements in a factor investing architecture are better obtained by adding new factor curves and model complexity or by improving portfolio construction through simple regime-conditioned allocation. Using a proprietary U.S. equity research panel, the study evaluates 198 factor curves built across 11 sectors, six factor definitions and three portfolio directions. The empirical design compares nine portfolio policies over a common weekly window from 2002-11-04 to 2025-01-13, including a raw alpha Baseline, a robust Hybrid reference, a defensive guard, interpretable regime rules, a supervised Logit selector, smooth sizing alternatives and a crash-transition overlay. The main historical finding is that a minimal regime-switching rule, referred to as V3 rule, delivered a stronger risk-adjusted compromise than either the raw Baseline or the Hybrid reference. V3 achieved 22.78% CAGR, 1.777 Sharpe and -14.61% maximum drawdown, compared with 24.36% CAGR, 1.632 Sharpe and -18.39% maximum drawdown for the Baseline and 21.45% CAGR, 1.731 Sharpe and -14.61% maximum drawdown for the Hybrid reference. The contribution is not a production trading strategy. It is a historical applied case study in research governance: when a raw factor engine already exists, the next useful gain may come from interpretable portfolio construction and regime conditioning rather than from more factors or more complex model layers.

Keywords: factor investing; regime switching; portfolio construction; factor timing; machine learning; backtest overfitting; research governance.

JEL Codes: G11, G12, G17, C52, C58.

Disclaimer

This paper is an independent research study. The results are based on historical simulations using a proprietary U.S. equity research panel. Reported returns are gross research returns and do not include transaction costs, borrow costs, market impact, taxes, slippage, or implementation constraints. The paper does not present a live trading strategy, investment recommendation, or production-ready portfolio. The V3 rule is disclosed only in semi-open form: the paper describes the economic logic, sleeves and high-level decision process, but does not disclose proprietary exact thresholds.

1. Introduction

Factor investing research often advances by expanding the factor zoo or by adding increasingly sophisticated timing models. Both routes can be valuable, but both also increase exposure to familiar model-risk problems: data mining, fragile specifications, unstable regime transfer, hidden implementation assumptions and the temptation to overinterpret a successful backtest.

This paper studies a more pragmatic question: when a factor platform already contains a raw return engine, does the next useful improvement come from more factor discovery or from better portfolio construction? The empirical setting is a proprietary U.S. equity research panel from which 198 factor curves are constructed across sectors, factor definitions and portfolio directions. Instead of asking which new factor should be added, the paper asks whether a simple regime rule can decide better when to rely on the raw alpha engine, when to use a diversified reference architecture and when to use a defensive guard.

The main result favors portfolio construction. In the common historical strategy window, the V3 rule does not beat the Baseline on raw CAGR. The Baseline remains the strongest raw return engine. However, V3 materially improves the drawdown profile of the Baseline and preserves the maximum drawdown of the Hybrid reference while improving both CAGR and Sharpe. This makes V3 the most interesting research candidate in the current evidence set.

The paper is deliberately conservative in its claims. The results are gross historical research returns, not live performance. The data are proprietary, not fully externally replicable from the paper alone. Promotion gates based on freshness, deflated Sharpe, probability of backtest overfitting and forward revalidation block any production claim. For this reason, the paper should be read as an applied research case study and as an example of research governance, not as an investable strategy specification.

2. Related Literature

This study connects to four streams of recent factor-investing research. First, it relates to machine-learning approaches to return prediction and market timing. AQR Portfolio Solutions Group (2024) argues that more complex models can detect nonlinearities in expected returns, but also emphasizes that gains in financial prediction are generally evolutionary rather than revolutionary. The Logit and smooth-sizing alternatives in this paper are included in that spirit: they are useful benchmarks for whether a more complex model layer improves a factor architecture, not presumed winners.

Second, the paper relates to dynamic factor allocation and regime-switching models. Shu and Mulvey (2024) study factor allocation through regime-switching signals and a Black-Litterman allocation framework. Northern Trust Asset Management (2025) also studies factor investing during stressed economic states and uses regime probabilities to adjust multi-factor weights. The present paper is more modest and more applied: it does not claim a general regime model, but asks whether an interpretable regime rule over existing sleeves can improve a historical factor-curve architecture.

Third, the paper connects to research-governance and causal-hygiene concerns in factor investing. Lopez de Prado and Zoonekynd (2025) emphasize that factor investing can suffer from causal neglect and factor mirages. Sadeghi, Gopal and Fesanghary (2024) further emphasize the difficulty of causal discovery in nonstationary financial time series. These concerns motivate the use of rejection logs, promotion gates and explicit limitations in the present study.

Fourth, the paper connects to portfolio construction and sector-aware factor design. Ehsani, Harvey and Li (2023) show that the effect of sector neutrality can differ across long-short and long-only settings. Bryzgalova, DeMiguel, Li and Pelger (2022) propose incorporating economic targets into factor estimation, reinforcing the broader point that economically meaningful objectives can matter as much as purely statistical fit. In the present study, the relevant objective is not only finding factors, but managing the allocation among factor-curve portfolios under changing regimes.

3. Data and Factor-Curve Universe

The empirical universe is a proprietary U.S. equity research panel. Internally, the source artifacts refer to local processed research panels and factor-curve outputs, but those internal names are not required for the external interpretation of the study. The article-ready sample includes weekly returns for the compared portfolio policies over the common strategy window from 2002-11-04 to 2025-01-13.

A factor curve is defined as a historical return series generated from a rule of the form sector x factor definition x portfolio direction. The universe contains 198 factor curves, equal to 11 sectors, six factor definitions and three portfolio directions. The directions are top_10, bottom_10 and long_short_10. The sectors are Basic Materials, Communication Services, Consumer Cyclical, Consumer Defensive, Energy, Financial Services, Healthcare, Industrials, Real Estate, Technology and Utilities. The factor definitions are defensive_compound, distress_avoidance, Piotroski_value_health, price_action_recovery, quality_cash_profitability and value_quality.

The raw factor-curve coverage spans 1985-02-01 to 2025-03-07, while the comparable weekly strategy window used in the main results spans 2002-11-04 to 2025-01-13. The latest local data date is 2025-03-07. No data refresh or new forward validation was performed for this manuscript version.

Table 1. Factor-curve universe summary

Item	Value
Factor curves	198
Sectors	11
Factor definitions	6
Portfolio directions	3
Raw curve coverage	1985-02-01 to 2025-03-07
Comparable strategy window	2002-11-04 to 2025-01-13
Latest local data date	2025-03-07
Strategy frequency	Weekly
Article weekly return rows	10,314

4. Portfolio Policies

The study compares nine portfolio policies. The Baseline is the raw alpha engine: it allocates to attractive factor-curve opportunities subject to a 45% sector cap. The Current Guard is a defensive shadow control that cuts risky exposure when stress conditions are active. The Hybrid reference combines guard behavior with a hierarchical risk parity sleeve and serves as the official historical reference architecture. Rule v1 is a simple interpretable meta-policy and serves as a benchmark against machine learning and V3.

The V3 rule is a minimal regime-switching policy over three sleeves: Baseline, Hybrid and Current Guard. The public disclosure is semi-open. The paper describes the sleeves, economic motivation and high-level pseudocode, but does not disclose exact proprietary thresholds. Over the sample, V3 allocates to the Baseline sleeve in 58.03% of weeks, to the Hybrid sleeve in 40.92% of weeks and to the Current Guard sleeve in 1.05% of weeks. This confirms that V3 is not primarily a cash-like or purely defensive rule. It is a switching policy that mostly alternates between the raw factor engine and the robust Hybrid reference, with rare use of the defensive guard.

The Logit policy tests whether supervised machine learning improves the sleeve-selection problem. The V2 ridge and V2 soft policies test whether smooth sizing beats a harder regime decision. The Crash Transition Guard adds an extra defensive overlay on top of V3. HRP standalone is included as a rejected path: it reduced risk but gave up too much return and Sharpe to become the primary engine.

Table 2. V3 sleeve attribution

Sleeve	Average weight	Weeks active	Pct weeks active	Contribution share
Baseline sleeve	58.03%	665	58.03%	62.23%
Current Guard sleeve	1.05%	12	1.05%	2.16%
Hybrid sleeve	40.92%	469	40.92%	35.61%

Semi-open V3 pseudocode

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For each weekly decision date:
  observe only information available at or before the feature date
  compute regime, stress, breadth and recent track-quality evidence

  if adverse regime evidence is materially elevated:
    allocate to Current Guard or Hybrid depending on defensive severity
  else if raw factor engine evidence is strong and drawdown evidence is acceptable:
    allocate to Baseline
  else:
    allocate to Hybrid

  record selected sleeve and weekly gross research return
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5. Research Design and Validation Metrics

All strategies are compared over the same weekly window. The main performance statistics are CAGR, annualized volatility, Sharpe ratio, maximum drawdown, Calmar ratio, hit rate, weekly 5% CVaR, worst week, average risky exposure and annualized allocation turnover. These statistics are computed on gross research returns.

The research process also includes internal promotion gates. These gates are not presented as definitive academic proof, but as governance checks. They include data freshness, reproducible outputs, use of a common OOS-style strategy window, improvement relative to the reference Sharpe, drawdown non-degradation, deflated Sharpe probability, probability of backtest overfitting, operational turnover and forward revalidation. V3 passes the internal research-candidate role but fails official promotion because of stale data, weak deflated-Sharpe evidence, PBO concerns and absence of forward revalidation.

The practical consequence is important. The manuscript can state that V3 is the best current research candidate in this historical evidence set. It should not state that V3 is an implementable, live or production-ready strategy.

6. Main Results

The core comparison is Baseline versus Hybrid versus V3. The Baseline has the highest CAGR, 24.36%, but also the largest maximum drawdown, -18.39%, and a lower Sharpe ratio of 1.632. The Hybrid reference lowers maximum drawdown to -14.61% and improves Sharpe to 1.731, but with a lower CAGR of 21.45%. V3 reaches 22.78% CAGR, 1.777 Sharpe and -14.61% maximum drawdown.

The result should be interpreted as a risk-adjusted improvement, not as a pure return improvement. V3 gives up 1.58 percentage points of CAGR relative to the Baseline, but improves maximum drawdown by 3.79 percentage points and improves Sharpe by 0.145. Relative to the Hybrid, V3 improves CAGR by 1.34 percentage points and Sharpe by 0.046 without worsening maximum drawdown.

This is the central empirical result of the paper: in this factor-curve architecture, a simple regime rule achieved a better historical compromise than adding further layers of complexity.

Table 3. Core comparison: Baseline, Hybrid and V3

Policy	CAGR	Sharpe	Max drawdown	Calmar	Interpretation
Baseline	24.36%	1.632	-18.39%	1.324	Raw alpha control; highest CAGR with worse drawdown.
Hybrid	21.45%	1.731	-14.61%	1.468	Official historical reference; robust defensive sleeve architecture.
V3 rule	22.78%	1.777	-14.61%	1.560	Best research candidate; simple regime rule improved Sharpe without worsening max

Policy	CAGR	Sharpe	Max drawdown	Calmar	Interpretation
					drawdown.

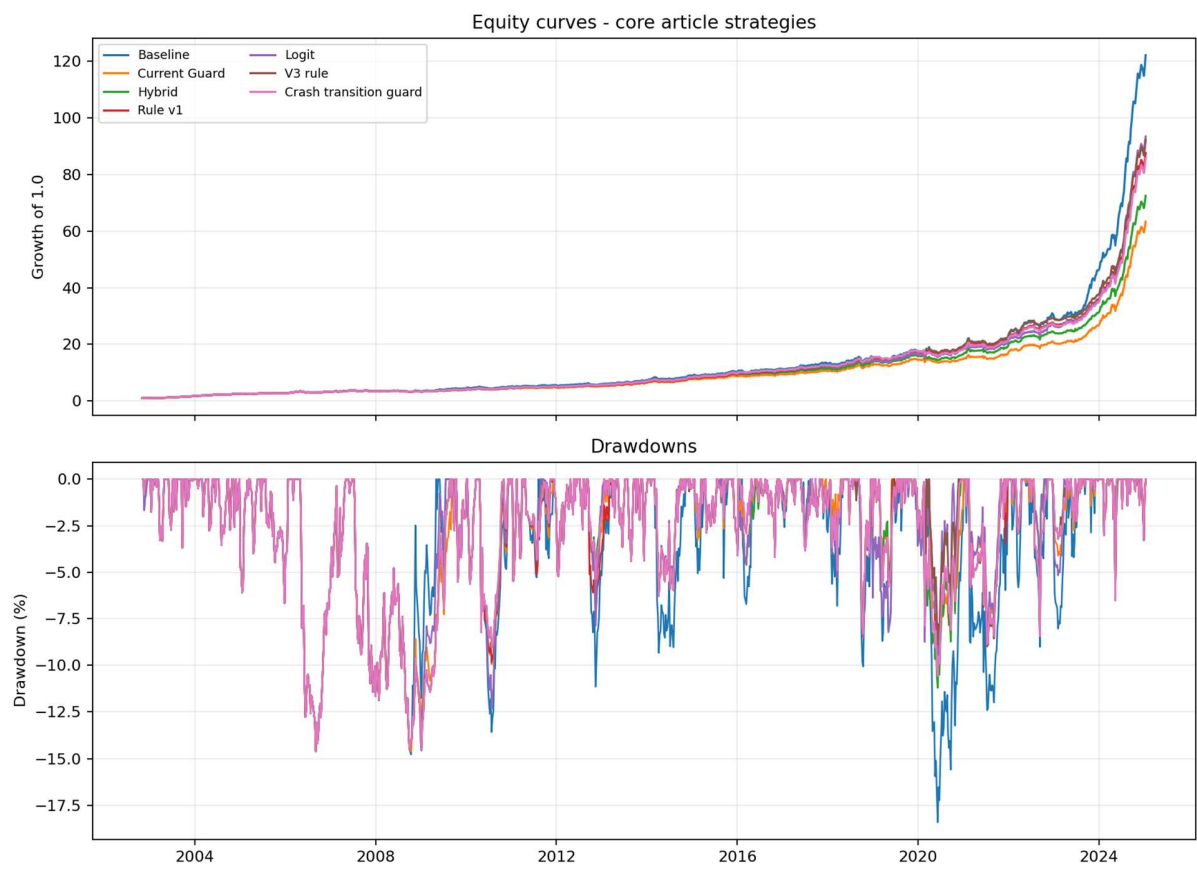


Figure 1. Core equity and drawdown comparison. V3 captures much of the Baseline return engine while preserving a drawdown profile closer to the Hybrid reference.

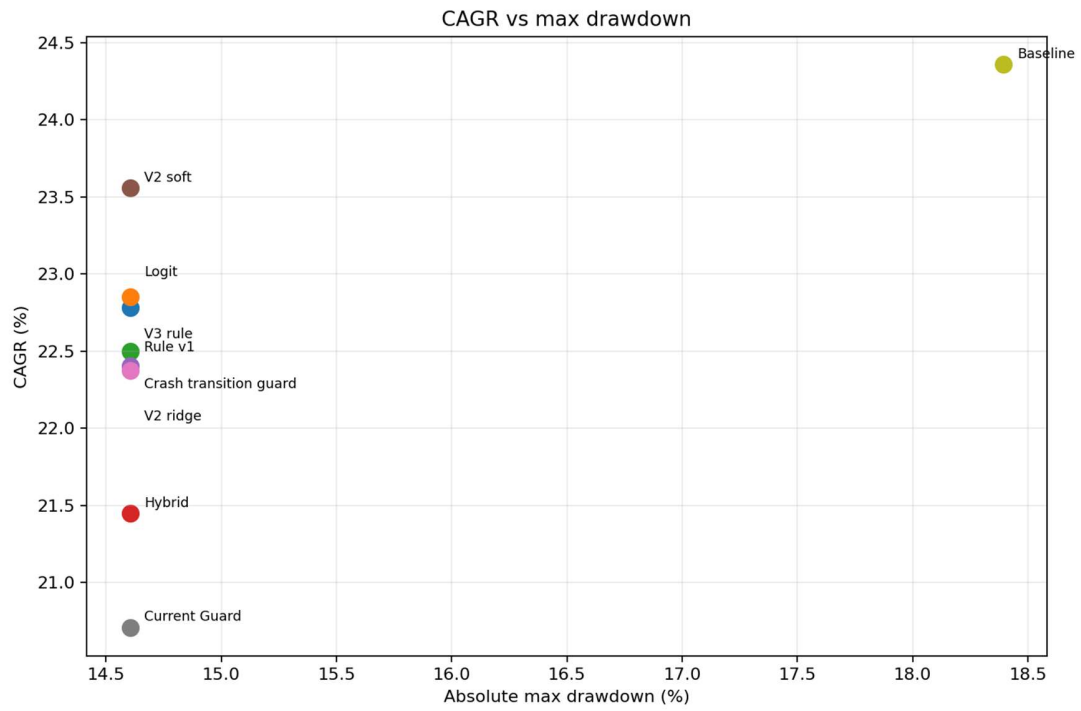


Figure 2. CAGR versus maximum drawdown. Baseline is the strongest raw return engine, while V3 is the more attractive risk-adjusted compromise.

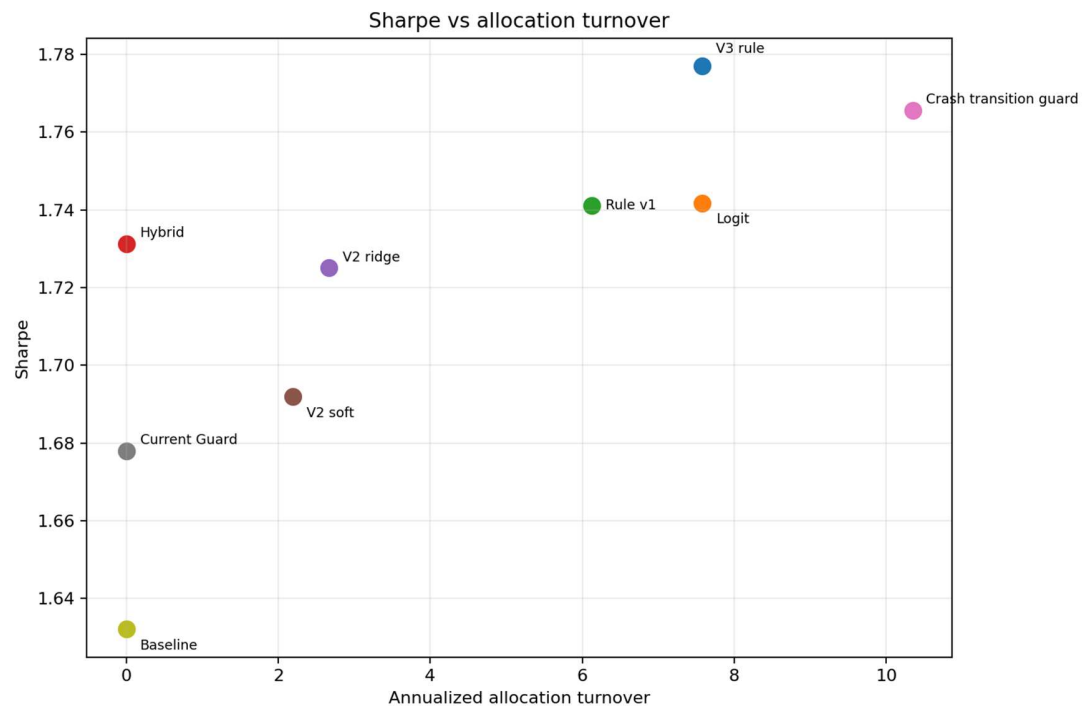


Figure 3. Sharpe versus allocation turnover. Higher turnover and added overlays do not necessarily improve the historical risk-adjusted frontier.

Table 4. All strategy results, gross historical research returns

Policy	CAGR	Sharpe	Max drawdown	Calmar	Weekly CVaR 5%	Alloc. turnover	DSR prob.	PBO below median	Role
V3 rule	22.78%	1.777	-14.61%	1.560	-3.22%	7.58	48.10%	32.86%	primary research candidate after refresh
Logit	22.85%	1.742	-14.61%	1.565	-3.31%	7.58	47.42%	10.00%	secondary research diagnostic
Rule v1	22.50%	1.741	-14.61%	1.540	-3.32%	6.13	47.81%	10.00%	secondary research diagnostic
Hybrid	21.45%	1.731	-14.61%	1.468	-3.19%	0.00	34.37%	22.73%	official reference now
V2 ridge	22.40%	1.725	-14.61%	1.534	-3.33%	2.66	44.42%	52.86%	research diagnostic do not promote
V2 soft	23.56%	1.692	-14.61%	1.613	-3.57%	2.19	39.59%	52.86%	research diagnostic do not promote
Crash transition guard	22.37%	1.766	-14.61%	1.532	-3.22%	10.35	47.59%	44.29%	failed overlay do not promote
Current Guard	20.71%	1.678	-14.61%	1.418	-3.21%	0.00	46.18%	20.91%	defensive shadow control
Baseline	24.36%	1.632	-18.39%	1.324	-3.88%	0.00	27.11%	20.91%	alpha shadow control

Note: DSR and PBO are internal robustness diagnostics. Allocation turnover is research allocation turnover, not an implementation-cost estimate.

7. Stress and Subperiod Evidence

The stress-bucket evidence strengthens the interpretation of V3 as a compromise between raw alpha and defense. In stress periods, the Baseline still has the highest CAGR at 23.06%, but this comes with a -17.74% maximum drawdown and -4.51% weekly CVaR. V3 has lower stress-period CAGR at 17.88%, but with a -8.99% maximum drawdown and -2.38% weekly CVaR. In other words, the raw engine remains powerful but more fragile; V3 gives up part of the raw return to improve downside behavior.

The decade-level evidence shows that V3 is not driven by one obvious subperiod only. In the 2000s, V3 and Hybrid are close on Sharpe and drawdown, while the Baseline has a slightly higher CAGR but similar drawdown. In the 2010s, V3 improves the Sharpe and drawdown trade-off relative to the Baseline. In the 2020s, the Baseline again has the highest CAGR, but also the worst drawdown among the three core strategies.

These results do not replace forward validation, but they help explain the economic role of the rule: V3 acts as an allocator between raw alpha and robustness, rather than as a new factor discovery engine.

Table 5. Stress-bucket results

Policy	CAGR	Sharpe	Max drawdown	Weekly CVaR 5%	Worst week
V3 rule	17.88%	1.728	-8.99%	-2.38%	-5.25%
Hybrid	13.59%	1.588	-9.17%	-2.24%	-5.25%
Crash transition guard	16.55%	1.698	-8.98%	-2.36%	-5.25%
Current Guard	11.27%	1.339	-9.24%	-2.27%	-4.42%
Baseline	23.06%	1.341	-17.74%	-4.51%	-8.74%

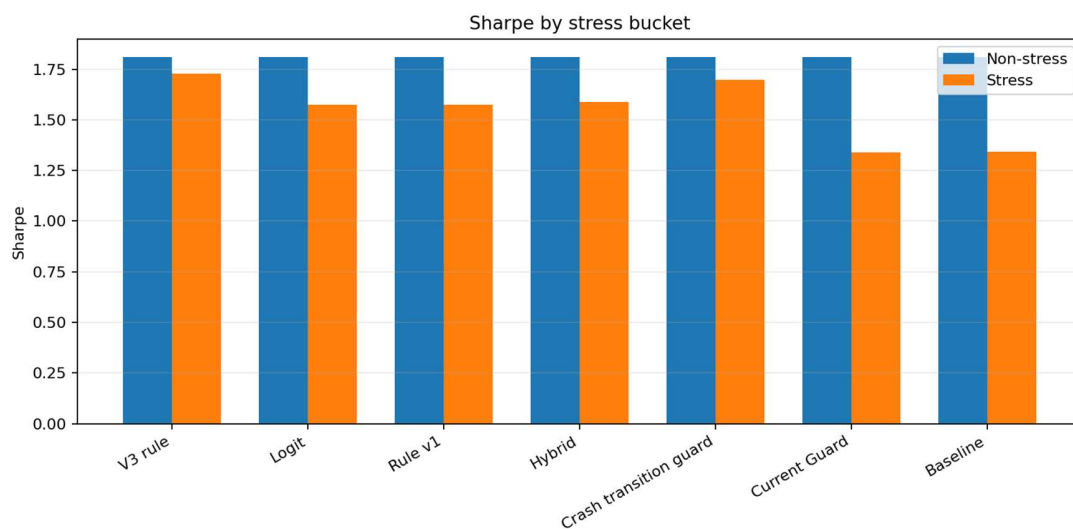


Figure 4. Stress-bucket Sharpe. V3 improves downside behavior relative to Baseline while retaining meaningful return exposure.

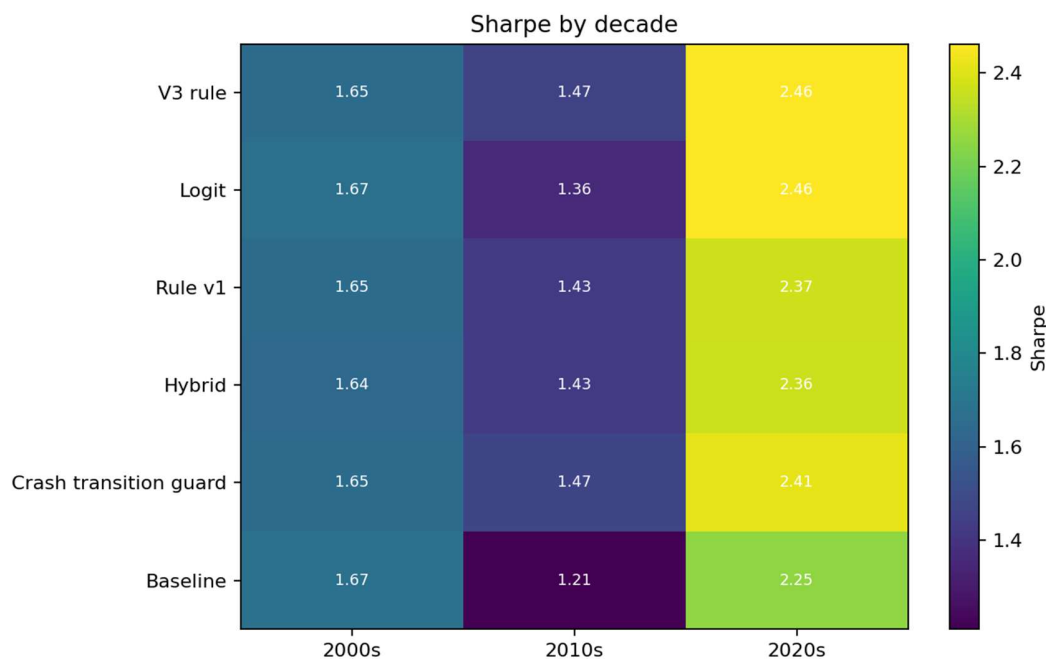


Figure 5. Sharpe by decade. Historical subperiod stability should be interpreted as supporting evidence, not a substitute for forward validation.

Table 6. Decade-level evidence for core strategies

Decade	Policy	CAGR	Sharpe	Max drawdown	Weeks
2000s	Baseline	23.60%	1.675	-14.77%	365
2000s	Hybrid	21.53%	1.639	-14.61%	365
2000s	V3 rule	21.86%	1.654	-14.61%	365
2010s	Baseline	14.78%	1.210	-13.58%	518
2010s	Hybrid	14.99%	1.435	-9.04%	518
2010s	V3 rule	15.82%	1.465	-9.04%	518
2020s	Baseline	46.85%	2.252	-18.39%	263
2020s	Hybrid	35.12%	2.365	-11.10%	263
2020s	V3 rule	39.21%	2.462	-9.00%	263

8. Complexity Audit and Negative Results

A central contribution of the paper is that it does not hide rejected paths. The project tested several plausible improvements: adding new factor curves, HRP as a standalone allocator, supervised meta-selection, smooth utility sizing and an additional crash-transition overlay. None dominated the final risk-adjusted frontier after governance checks.

The new-factor satellite path produced only a small CAGR gain and did not preserve the drawdown and Sharpe acceptance gates. HRP standalone reduced drawdown but gave up too much CAGR and Sharpe, making it more useful as a sleeve than as the core engine. Logit was competitive, but did not provide enough robustness premium over simpler rules to justify making supervised machine learning the main claim. V2 smooth sizing diluted the regime decision and had weaker PBO/DSR evidence. Crash Transition Guard over-protected: it reduced CAGR and Sharpe, increased turnover and failed to improve global maximum drawdown.

The rejection log supports the title claim. The outcome was not simply that V3 looked good after an unconstrained search. Rather, multiple natural extensions were documented and rejected. This makes the paper as much about research governance as about a specific historical rule.

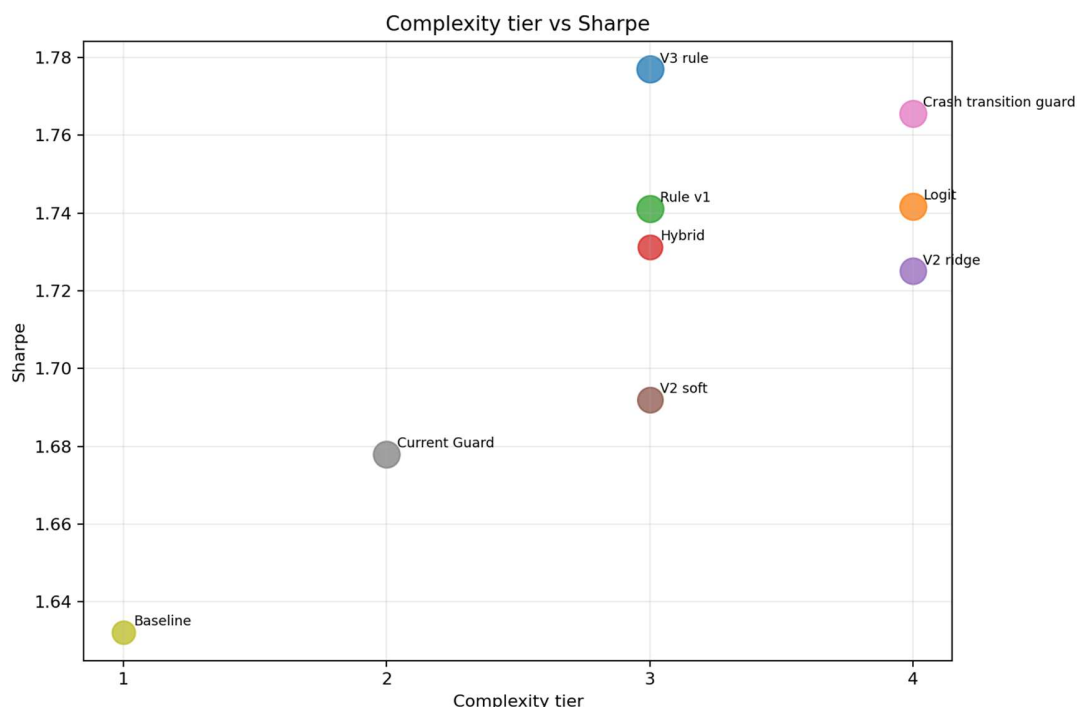


Figure 6. Model complexity versus Sharpe. More complex strategies did not dominate the simpler V3 regime rule after considering robustness and promotion gates.

Table 7. Rejected and diagnostic paths

Branch	Outcome	Reason not promoted	Paper use
GC004 triple-barrier target trading labels	reject_or_reformulate	Take-profit label lift did not translate into mark-to-market portfolio excess return.	Methodological negative result: target labels are not portfolio returns.
New Factor Selective Integration v3	do_not_integrate	Small CAGR gain did not preserve drawdown/Sharpe acceptance gates.	Supports 'not more factors' thesis.
HRP / Cluster Allocator v1	research_only_sleeve	Reduced drawdown, but gave up too much CAGR and Sharpe.	Supports sleeve-not-engine conclusion.
Meta-Policy Selector v2	do_not_promote	Ridge/sizing diluted the regime decision and had weak PBO/DSR.	Shows more model sophistication did not improve frontier.
Crash Transition Guard v1	do_not_promote	Reduced CAGR/Sharpe and did not improve max drawdown; PBO worsened.	Shows extra overlays can over-protect.

9. Research Governance and Limitations

The strongest limitation is that the results are historical gross research returns. The reported performance excludes transaction costs, borrow costs, market impact, taxes, slippage and operational implementation constraints. Long-short factor curves may be materially affected by borrow availability, financing costs and shorting constraints. Allocation turnover should therefore be interpreted as research allocation turnover, not as a full implementation-cost estimate.

The proprietary data panel limits external replicability. The paper can describe the factor-curve construction and provide article-ready outputs, but it cannot make the entire local research panel public. The data are also stale relative to a production decision, and no new forward validation was performed for this version.

The V3 rule is disclosed semi-openly. The economic logic, sleeves, timing convention and average exposures are disclosed, but exact proprietary thresholds are not. This protects research IP but limits full third-party replication.

The internal promotion gates also counsel caution. V3 passes a research-candidate role, but fails official promotion because of data freshness, deflated-Sharpe, PBO and forward-revalidation gates. These gates are not a reason to discard the historical finding. They are the reason to state the finding carefully.

Table 8. Selected promotion status and failed gates

Policy	Research pass	Official pass	Failed gates
V3 rule	Pass	Fail	G0_data_freshness G5_deflated_sharpe G6_pbo G8_forward_revalidation
Crash transition guard	Fail	Fail	G0_data_freshness G5_deflated_sharpe G6_pbo G7_operational_turnover G8_forward_revalidation
Hybrid	Fail	Fail	G0_data_freshness G3_reference_sharpe_margin G5_deflated_sharpe G6_pbo G8_forward_revalidation
Baseline	Fail	Fail	G0_data_freshness G3_reference_sharpe_margin G4_drawdown_non_degradation G5_deflated_sharpe G6_pbo G8_forward_revalidation

Note: V3 is a research candidate, not an official production promotion.

10. Conclusion

This paper documents a historical empirical case study in factor-curve portfolio construction. The main finding is that, in a proprietary U.S. equity research panel, the most useful improvement did not come from adding more factors or adding more complex overlays. It came from a minimal, interpretable regime rule that switched among an existing raw alpha engine, a robust Hybrid reference and a defensive guard.

The V3 rule should not be interpreted as a final strategy. It is a research candidate. Its historical contribution is to show that, when a raw factor engine already exists, the next useful research frontier may be deciding when to carry, reduce or diversify risk. The broader lesson is therefore practical: before expanding the factor zoo or adding model layers, test whether simple portfolio construction and regime conditioning already capture the most valuable marginal improvement.

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